API conventions

English version of ../../api/conventions.md.

Specification: backend/openapi.json — generated (make openapi), checked in CI (make openapi-check). Interactive documentation runs on a live instance: /api/v1/docs.

Base rules

Rule	Detail
Prefix	/api/v1 — set in code (backend/app/main.py), no environment variable
Format	JSON
Authentication	Bearer JWT (backend/auth), cookie session for the browser
Time	ISO 8601 with a timezone; calculations are pinned to AEZA_AS_OF
Money	integer minor units plus a currency, never float
Errors	FastAPI {"detail": ...} plus domain codes from backend/app/domain/*

Key route groups

Group	Purpose	Access
/auth/*	sign-in, refresh, sign-out	—
/analytics/*	aggregates, risk, revenue at risk	analyst role
/customers/*	accounts and services	analyst role
/insights/*	AI insights	analyst role
/actions/*	suggested actions and scenarios	analyst role
/source/snapshot	snapshot ingestion	AEZA_SOURCE_INGEST_TOKEN
/export/*	exports	analyst role
/ai/*, /model/*	assistant and model state	analyst role
/telegram/*	Mini App and bot	Telegram signature
/health, /metrics	operations	AEZA_METRICS_TOKEN for metrics

Idempotency and limits

- Snapshot ingestion (PUT /api/v1/source/snapshot) is a replacement: repeating the same body gives the same result.
- Size and timeout limits — AEZA_AI_TIMEOUT_SECONDS, AEZA_OLLAMA_TIMEOUT_SECONDS, upload limits in backend/app/api/routes_source.py.
- Application-level rate limiting is not implemented: rely on a proxy or gateway. That is a known limitation, not an oversight — see operations/deployment.md.

Errors

Code	Meaning
400	invalid input (domain `InvalidInput`)
401	missing or expired token
403	no role or no access to the object
404	the object does not exist
409	conflict (`ConflictError`, for example a repeated action)
422	request schema error
503	the analytics domain is unavailable (`UnavailableAnalytics`) — an honest refusal, not empty data

How to change the API

1. Change the route or schema; 2. `make openapi`; 3. update the client in `frontend/src`;
2. `python backend/scripts/check_docs.py` (documentation mentions only existing routes);
3. `make verify`.